

1. The sum of the first k terms of a series S is $3k^2 + 5k$. Which one of the following is correct?

- (a) The terms of S form an arithmetic progression with common difference 14.
- (b) The terms of S form an arithmetic progression with common difference 6.
- (c) The terms of S form a geometric progression with common ratio $10/7$.
- (d) The terms of S form a geometric progression with common ratio $11/4$.

2. The sum of the first 8 terms of a GP is five times the sum of its first 4 terms. If $r \neq 1$ is the common ratio, then what is the number of possible real values of r ?

- (a) One
- (b) Two
- (c) Three
- (d) More than three

3. If one root of the equation $x^2 - kx + k = 0$ exceeds the other by $2\sqrt{3}$, then which one of the following is a value of k ?

- (a) 3 (b) 6
- (c) 9 (d) 12

4. If $x + \frac{5}{y} = 4$ and $y + \frac{5}{x} = -4$, then what is $(x + y)$ equal to?

- (a) 0 (b) 1
- (c) 4 (d) 5

5. If 5th, 7th and 13th terms of an AP are in GP, then what is the ratio of its first term to its common difference?

- (a) -3
- (b) -2
- (c) 2
- (d) 3

6. If $p, 1, q$ are in AP and $p, 2, q$ are in GP, then which of the following statements is/are correct?

- I. $p, 4, q$ are in HP.
- II. $(1/p), 1/4, (1/q)$ are in AP.

Select the answer using the code given below.

- (a) I only
- (b) II only
- (c) Both I and II
- (d) Neither I nor II

7. If $x = (1111)_2$, $y = (1001)_2$ and $z = (110)_2$, then what is $x^3 - y^3 - z^3 - 3xyz$ equal to?

- (a) $(1111001)_2$
- (b) $(1001111)_2$
- (c) $(1)_2$
- (d) $(0)_2$

8. यदि

$$\Delta = \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix}$$

और A, B, C, D, G अवयवों a, b, c, d, g के क्रमशः सहखंड हैं, तो $bB + cC - dD - gG$ किसके बराबर है?

- (a) 0
(b) 1
(c) Δ
(d) $-\Delta$

9. सारणिक

$$\Delta = \begin{vmatrix} k(k+2) & 2k+1 & 1 \\ 2k+1 & k+2 & 1 \\ 3 & 3 & 1 \end{vmatrix}$$

के संदर्भ में निम्नलिखित कथनों पर विचार कीजिए :

- I. Δ धनात्मक है, यदि $k > 0$.
II. Δ ऋणात्मक है, यदि $k < 0$.
III. Δ शून्य है, यदि $k = 0$.

उपर्युक्त कथनों में से कितना/कितने सही है/हैं?

- (a) कोई भी नहीं
(b) एक
(c) दो
(d) सभी तीन

10. यदि

$$\begin{vmatrix} 2 & 3+i & -1 \\ 3-i & 0 & i-1 \\ -1 & -1-i & 1 \end{vmatrix} = A + iB$$

जहाँ $i = \sqrt{-1}$, तो $A + B$ किसके बराबर है?

- (a) -10
(b) -6
(c) 0
(d) 6

11. यदि $A^2 + B^2 + C^2 = 0$, तो

$$\begin{vmatrix} 1 & \cos C & \cos B \\ \cos C & 1 & \cos A \\ \cos B & \cos A & 1 \end{vmatrix}$$

का मान क्या है?

- (a) -1
(b) 0
(c) 1
(d) 2

12. यदि ω एक (unity) का अवास्तविक घनमूल है, तो समीकरण

$$\begin{vmatrix} x+1 & \omega & \omega^2 \\ \omega & x+\omega^2 & 1 \\ \omega^2 & 1 & x+\omega \end{vmatrix} = 0$$

का एक मूल (root) क्या है?

- (a) $x = 0$
(b) $x = 1$
(c) $x = \omega$
(d) $x = \omega^2$

13. $\left(\frac{\sqrt{3}+i}{\sqrt{3}-i}\right)^3$ किसके बराबर है?

- (a) -1
(b) 0
(c) 1
(d) 3

14. यदि $x^2 - x + 1 = 0$, तो

$$\left(x - \frac{1}{x}\right)^2 + \left(x - \frac{1}{x}\right)^4 + \left(x - \frac{1}{x}\right)^8$$

किसके बराबर है?

- (a) 81
(b) 85
(c) 87
(d) 90